🐍 Python Learning Journey – Engineering Student

> Documenting my Python learning path as an engineering student – from fundamentals to real-world engineering applications.

🧑‍💻 About Me

Hi! I'm an engineering student learning Python to solve real-world problems – from data analysis and automation to simulations and embedded systems. This repo is my personal notebook, code collection, and progress tracker.

📁 Repository Structure

```
python-learning/
│
├── 01_basics/
│   ├── variables_and_datatypes.py
│   ├── control_flow.py
│   ├── functions.py
│   └── README.md
│
├── 02_data_structures/
│   ├── lists_tuples.py
│   ├── dicts_sets.py
│   ├── comprehensions.py
│   └── README.md
│
├── 03_oop/
│   ├── classes_objects.py
│   ├── inheritance.py
│   ├── encapsulation.py
│   └── README.md
│
├── 04_file_handling/
│   ├── read_write.py
│   ├── csv_json.py
│   └── README.md
│
├── 05_libraries/
│   ├── numpy_basics.py
│   ├── pandas_basics.py
│   ├── matplotlib_plots.py
│   └── README.md
│
├── 06_projects/
│   ├── calculator/
│   ├── grade_analyzer/
│   ├── data_visualizer/
│   └── README.md
│
└── resources/
    └── cheatsheets/
```

🗺️ Learning Roadmap

Phase	Topic	Status
1	Python Basics (variables, loops, functions)	✅ Done
2	Data Structures (lists, dicts, sets)	✅ Done
3	Object-Oriented Programming	🔄 In Progress

4	File Handling & I/O	🔜 Upcoming
5	NumPy & Pandas	🔜 Upcoming
6	Matplotlib & Data Visualization	🔜 Upcoming
7	Engineering Projects	🔜 Upcoming

📝 Weekly Log

Week 1 — Python Basics
- Variables, data types, type casting
- `if/elif/else`, `for`, `while` loops
- Functions, arguments, `*args`, `**kwargs`

Week 2 — Data Structures
- Lists and tuples (mutability, slicing)
- Dictionaries and sets
- List/dict comprehensions

Week 3 — OOP *(in progress)*
- Classes and objects
- `__init__`, instance vs class variables
- Inheritance and method overriding

🛠️ Projects

🔢 Calculator
A simple CLI calculator with support for basic arithmetic operations.
→ [`06_projects/calculator/`](./06_projects/calculator/)

📊 Grade Analyzer
Reads student grades from a CSV and outputs statistics and a grade distribution plot.
→ [`06_projects/grade_analyzer/`](./06_projects/grade_analyzer/)

📈 Data Visualizer *(coming soon)*
Visualizes engineering datasets using Matplotlib and Pandas.

💡 Key Learnings & Notes

- Python uses **indentation** to define blocks — no braces needed.
- `list.append()` vs `list.extend()` — one adds an element, the other merges iterables.
- Always use `if __name__ == "__main__":` to make scripts importable as modules.
- NumPy arrays are **much faster** than Python lists for numerical computation.

🔧 Setup & Requirements

```bash
# Clone the repo
git clone https://github.com/your-username/python-learning.git
cd python-learning

# Create a virtual environment
python -m venv venv
source venv/bin/activate  # Windows: venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt
```

requirements.txt
```
numpy
```

pandas
matplotlib
jupyter
```

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## 📚 Resources I'm Using

- [Python Official Docs](https://docs.python.org/3/)
- [CS50P - Harvard's Python Course](https://cs50.harvard.edu/python/)
- [Automate the Boring Stuff with Python](https://automatetheboringstuff.com/)
- [Real Python](https://realpython.com/)
- [NumPy Documentation](https://numpy.org/doc/)

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## 📌 Goals

- [ ] Complete all 6 learning phases
- [ ] Build 3 functional mini-projects
- [ ] Write a Python script to automate a real engineering task
- [ ] Explore Python in Arduino/Raspberry Pi context
- [ ] Contribute to an open-source Python project

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## 📬 Connect

Feel free to open an issue, suggest resources, or share feedback!

> *"Code is like humor. When you have to explain it, it's bad." – Cory House*

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⭐ Star this repo if you find it helpful — it keeps me motivated!
```